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Vinay Ram Gazula

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An organised and highly motivated individual who always does at the top of his capabilities, seeking an admission for Master's program at your University where I can build a valuable skill set and launch my career.

Education

Bachelor of Technology in Mechanical Engineering **2017-2021**
SRM University, AP CGPA: 8.51
Awarded with merit scholarship of 50% tuition fee for 4 years.

Board of Intermediate Education Andhra Pradesh, BIEAP (12th) **2015-2017**
Narayana Jr. College Percentage: 95.1%

Central Board of Secondary Education, CBSE (10th) **2014-2015**
Sri Chaitanya Techno School CGPA: 9.4

Skills

Programming Languages	Python, C/C++, Matlab, Bash Shell Scripting (Linux)
Developer Tools	Git/GitHub, Spyder, KDE Advanced Text Editor (KATE), VS Code
Documentation Tools	MS Office, LaTeX, reStructuredText (Sphinx)
Frameworks & Libraries	OP2 Framework, Pandas, yfinance, NumPy, SymPy, Matplotlib, Tensorflow
CAD & CFD Softwares	Solidworks, CATIA, Tecplot 360, Gmsh, ANSYS Fluent, Paraview
Operating Systems	Windows, Linux, MacOS

Experience

Research Intern **Sep 2021 - Feb 2022**
Under Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University, AP)

- Developed Perfect gas solver code from the existing CO_2 solver code.
- Worked on the development of solver code for gaseous reactions and viscous properties.
- Validated the solver for different flow configurations like Single ramp, Double ramp, etc,.
- Published a conference paper in International ISHMT-ASTFE Heat and Mass Transfer Conference.

Internship on CATIA **Jun - Aug 2020**
AP State Skill Development Corporation (APSSDC) *Online*

- Practical experince with Catia V5 software.
- Designed a Turbocharger assembly for automobiles using Catia.

Publications

- Vinay, G., Dinesh, D., Satya, P., "Accelerating a Hypersonic CO_2 Reaction Solver", ***Proceedings of the 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference December 17-20, 2021, IIT Madras, Chennai-600036, Tamil Nadu, India.*** *(Reference)*

Projects

Developing a Stock Trading Bot **Mar - Jul 2022**
Personal Project

- Designed and developed a fully automated stock trading bot using Python.
- The bot gathers any desired stock's historical price data, performs technical analysis using technical indicators and places orders to the stock broker using the broker API.

Accelerating a Hypersonic CO_2 Reaction Solver **Jan - May 2021**
B.Tech Final Year Project *Academic Project*

- A 2-D unstructured finite volume method based solver capable of simulating hypersonic flows in CO_2 rich Martian atmosphere with chemical kinetics for eight species consisting of $N_2, O_2, NO, O, N, CO_2, CO$ and C incorporated in it is developed using python.
- The solver is parallelized to run on single or multiple CPUs and GPUs in order to attain greater computational speeds using OP2 application programming interface.

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Design and Analysis of Heat Sink **Oct - Nov 2020**
Thermal Design of Electronic Equipment Course *Coursework Project*

- Designed and analysed a 2-D heat sink made of Aluminium and a layer of epoxy between the chip and heat sink with specified thermal properties as a thermal design solution for cooling a noval chip (CPU for example) using CFD-HT (Computational Fluid Dynamics and Heat Transfer) software ANSYS Fluent.

Flow through an Exhaust Manifold of 4-Stroke Diesel Engine **Feb - May 2020**
Computational Fluid Dynamics Course *Coursework Project*

- Designed an exhaust manifold for a V-8 engine in Solidworks and then imported this geometry into the Ansys workbench.
- Using Fluent, simulated the flow of the hot exhaust gases coming out from the engine and designed various diffuser models and studied the flow behavior for different diffuser models and submitted the final report for the optimal design.

Sub-Sonic Wind tunnel **Sept - Dec 2019**
Undergraduate Research Program (UROP) *Academic Project*

- Developed a Subsonic wind tunnel, which is designed to achieve 30m/s in the test section with expected low intensity turbulence level, making it available for researching in areas such as low speed aerodynamics and other such possibilities.

Certifications

Autodesk Fusion 360 Integrated CAD/CAM/CAE **Jan 2020**
Coursera Online
Introduction to Programming with MATLAB **May 2020**
Coursera Online
Robo Feast 2018 **Feb 2018**
Certificate of Participation
Participated in the 2 days National Level Workshop cum Championship on "*WiFi-Controlled Robotics*" organized by ***Revert Technology Pvt. Ltd. & EDC, IIT Roorkee.***